CSE STUDY HUB

Digital Forensics & Incident Response

UNIT 2 · Evidence Acquisition and Preservation

What this unit is about

This unit builds understanding of **Evidence Acquisition and Preservation**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Acquisition planning, authority, scope and documentation

2. Bit-stream/physical imaging vs logical acquisition

3. Live acquisition and dead-box acquisition; when to use each

4. Write blockers and read-only evidence handling

5. Disk imaging workflow and forensic image formats

6. Hashing with MD5/SHA family for integrity checking; limitations of MD5 for security

7. Pre- and post-acquisition hash verification

8. Order of volatility and capturing RAM, processes and network state

9. Chain-of-custody forms, evidence packaging, storage and preservation

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 2 NOTES · PART 1

Acquisition planning, authority, scope and documentation

Acquisition planning, authority, scope and documentation is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Bit-stream/physical imaging vs logical acquisition

Compare the alternatives in Bit-stream/physical imaging vs logical acquisition using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Live acquisition and dead-box acquisition; when to use each

Live acquisition and dead-box acquisition; when to use each is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Evidence Acquisition and Preservation, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 2 NOTES · PART 2

Write blockers and read-only evidence handling

Write blockers and read-only evidence handling is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Disk imaging workflow and forensic image formats

Present Disk imaging workflow and forensic image formats as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Hashing with MD5/SHA family for integrity checking; limitations of MD5 for security

A hash function maps a key to a table index. Collisions are handled by chaining or open addressing. Performance depends on a good hash function, load factor and resizing policy.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Evidence Acquisition and Preservation, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 2 NOTES · PART 3

Pre- and post-acquisition hash verification

A hash function maps a key to a table index. Collisions are handled by chaining or open addressing. Performance depends on a good hash function, load factor and resizing policy.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Order of volatility and capturing RAM, processes and network state

Present Order of volatility and capturing RAM, processes and network state as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Chain-of-custody forms, evidence packaging, storage and preservation

Chain-of-custody forms, evidence packaging, storage and preservation is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Evidence Acquisition and Preservation, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Acquisition planning, authority, scope and documentation** — know the definition, core working and one example.

• **Bit-stream/physical imaging vs logical acquisition** — know the definition, core working and one example.

• **Live acquisition and dead-box acquisition; when to use each** — know the definition, core working and one example.

• **Write blockers and read-only evidence handling** — know the definition, core working and one example.

• **Disk imaging workflow and forensic image formats** — know the definition, core working and one example.

• **Hashing with MD5/SHA family for integrity checking; limitations of MD5 for security** — know the definition, core working and one example.

• **Pre- and post-acquisition hash verification** — know the definition, core working and one example.

• **Order of volatility and capturing RAM, processes and network state** — know the definition, core working and one example.

• **Chain-of-custody forms, evidence packaging, storage and preservation** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Evidence Acquisition and Preservation. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Acquisition planning, authority, scope and documentation and explain its importance.

Q2. Explain the working/steps involved in Bit-stream/physical imaging vs logical acquisition.

Q3. Differentiate between two related concepts from Evidence Acquisition and Preservation.

Q4. Give one practical application of Chain-of-custody forms, evidence packaging, storage and preservation and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Evidence Acquisition and Preservation.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.